The Normative Pull of Prediction Markets on Our Beliefs and Credences

Korbinian Friedl

1 Introduction

The market price of a bet on “Zohran Mamdani will win the New York City mayoral election” on Polymarket (“New York City Mayoral Election” 2025) is currently (July 7 2025) 69¢ — nice.

What should this mean for our doxastic and credal attitudes towards the proposition that Mamdani will win? Is the market price being what it is a reason to believe it? Would it be a reason to believe it if it were higher? Is it a reason to believe a proposition like “The chance of Mamdani winning is 69%”? To set your credence to 69% (or to any other specific or unspecific value)?

These are the questions I will be addressing in this paper; or, more precisely, I will be discussing the general question(s): In the context of a prediction market, let a bet on a proposition q be an asset b_q that pays \$1 if q and \$0 if $\neg q$; and let p_q be the market price of such a bet in dollars. What—if anything—is wrong with me if my credence in q is different from p_q? And is there a threshold such that there is something wrong with me if I fail to believe q despite the price of a bet on q being above that threshold?

I will sketch a way of thinking about beliefs and credences—inspired by and building upon MacFarlane (2025)—and use it to argue for the following answers:

Sometimes, it can be wrong for one’s credence in q to deviate from the market price of b_q. One such case is an agent who believes in one form or another of the efficient market hypothesis (and believes that the available prediction markets for q are such that it applies), and also believes that they do not possess exclusive information of the relevant kind (where the relevant kind depends on the specific form of the EMH the agent believes).

Clearly, if the threshold view or the Lockean thesis (Jackson 2020, 2–4) about the (metaphysical and normative, respectively) relationship between belief and credence is true, this will directly entail that market prices of bets likewise exert a normative pull on an agent’s beliefs. But the view I will be using does not have this feature. On the contrary, I will use it to argue that it is almost never be the case that the market price of a bet on q is a on its own a reason in favour of believing (or disbelieving) q (or $\neg q$).

Before I give a brief summary of the arguments I will be making, a brief note on terminology: A lot of the argument of this paper turns on the distinction between two normativities, the normativity governing moves in the language game of giving and asking for reasons, and the normativity governing actions properly guided by decision theory (which, I will assume, are exactly those actions for which both doing them, and refraining from doing them, is permitted). I will call the first by the name of "reasonableness", and the second by the name of "rationality*". I hesitate to call it "rationality", because rationality may have more expansive requirements, or may apply also to other realms of human activity. Hence, it will be "rationality*" in this paper[1].

My first argument is in support of the positive conclusion that credences sometimes need to track market prices of bets:

1.0.1 Argument 1

1. An agent's credences are irrational* if they decision-theoretically recommend irrational* actions.
2. Sometimes, betting against the market is irrational*.
3. ∴ Sometimes, an agent's credences need to be such as to not recommend betting against the market.
4. Credences which differ from the market price of a bet on q recommend a bet against the market.
5. ∴ Sometimes, an agent's credence in q needs to be identical to the market price of a bet on q.

For the negative conclusion that it is almost never the case that the market price of a bet on q is on its own a reason to believe q, I give the following argument. Let t be some threshold for which we might think that $p_q > t$ gives an agent a decisive reason to believe q.

1.0.2 Argument 2

1. If $p_q > t$ is on its own a reason to believe q, then an agent who accepts $p_q > t$ should by default be willing to also accept q as a reason for further claims or attitudes.
2. If $p_q > t$ is the only evidence available to an agent, it will in may cases be wrong for her to accept q as a reason for further claims or attitudes.
3. Therefore, $p_q > t$ not on its own a reason to believe q.

The core of these arguments is based on the aforementioned view on the relationship between belief and credence based on MacFarlane (2025). It will be developed in section Section 2, which will also make clear why and how it motivates premises 1 and 4 in the first argument and premise 1 in the second. The remaining premise (premise 2) of argument 1 will be discussed in section Section 4.1. Premise 2 of

[1] As MacFarlane, who I am following here, writes—though he sticks with using the word „rationality“ even in the face of such considerations: " I am rejecting a long philosophical tradition of thinking of rationality as the faculty for apprehending and responding to reasons. I'm even going against etymology, which derives 'rational' from ratio, reason. But one kind of philosophical progress is the recognition that there are two distinct things which we confused before" (MacFarlane 2025, 857).

argument 2 will be argued for in section Section 3 .

2 Belief and Credence

I will be thinking about the concepts of "belief" and "credence" in a methodologically behaviourist way here, that is, as theoretical terms of psychology which are to be introduced in terms of vocabulary pertaining to overt behaviour[2]. As such, they need to earn their keep by helping us make sense of, explain and predict the behaviour of human beings. The question to ask, then (along the lines of MacFarlane (2025) who this analysis is based on), is: What talents do these concepts bring to the table, and what job should they be given in our psychological theories?

I will start in section Section 2.1 by showcasing the utility of the concept of "credence" in describing and explaining human behaviour. I will then move on to describe the two distinctive talents which the concept of "belief" has above and beyond what "credence" can do, and introducing the specific job which MacFarlane has in mind for it in within our theorising of the human being and it's actions in section Section 2.2. The result will be a form of belief-credence dualism, whose explanatory power, as well as perhaps counter-intuitive consequences will be showcased in applying it to the "Sleeping Beauty" problem in section Section 2.4.

Section Section 2.5 will raise the question of whether perhaps this dualism has pulled belief and credence so far apart as to make unintelligible how they interact. It will answer this question by turning to the practice of betting as a bridge between the two attitudes.

2.1 The Power of Credence

Why do we even need credence in our theoretical vocabulary? Isn't belief enough?

The following example from MacFarlane (2025) shows the limitations of belief in explaining human action:

> Sam is walking on a narrow board that spans a shallow ditch. Why is he doing that? Because his lunch is on the other side of the ditch, and he believes that by walking across the board, he will get there. His action is rational in light of his goals and beliefs, and thus we explain it. But now let's consider a similar case, but let the board cross a chasm one hundred meters deep. Now Sam does not cross. Why not? He still wants to get to the other side, and he still believes that by walking across the board, he will get there. Yet, if we make the gap deep enough, it ceases to be rational for him to cross. (MacFarlane 2025, 848)

[2] In this (and some other) ways, the analysis here bears resemblance to the one Wilfrid Sellars gives of "thoughts" as inner episodes in §§46-59 of "Empiricism and the Philosophy of Mind" (Sellars 1997).

In simple cases, it looks like beliefs and desires do explain (individual) behaviour; but as the example shows, the (binary) concept of belief is lacking in the quantitative structure needed to account for some of the relevant differences in the modified situation. Credence-based decision theory, on the other hand, is able to capture these excellently:

"Suppose Sam's utilities for the various possible outcomes are as follows:

Outcome	Utility
Crosses successfully and gets lunch	1
Stays put and skips lunch	-1
Falls into shallow ditch	-1
Falls into deep chasm	-1000

Then, if Sam's credence that he will get across the board without falling is 0.99, the expected utility of crossing is 0.98 in 'Shallow' and -9.01 in 'Deep', while the expected utility of not crossing is -1 in both cases." (MacFarlane 2025, 848)

Consequently, what the role of credence in or psychological theorising should be, is the following: We describe an agent as having a certain set of credences iff those credences in fact decision-theoretically rationalise and explain their actions: Credences are those internal states which disposes one to make the decision which maximises the utility one expects based them.

2.2 Belief's Distinctive Talent

So the concept of "credence" proves its worth in rationalising and explaining those actions which are properly seen as governed by decision theory. Do we additionally need a concept of "belief"? Or can credence do everything we need in cognitive psychology?

Perhaps saying "A believes p" is simply an imprecise way of saying that A has a high credence in p? If this were so, it would explain why "belief" continues to be used in everyday language. But it wouldn't vindicate it as a scientific term. As MacFarlane writes, "just as we don't do physics with vague concepts like tall, we won't do psychology with belief" (MacFarlane 2025, 849).

Similarly unsatisfying are accounts on which belief serves to simplify cognition and thereby make it computationally tractable (such as Staffel (2019), Ross and Schroeder (2014)). If, as Ross and Schroeder (2014) write, belief in a set of propositions is "a defeasible or default disposition to treat them as true in our reasoning" (Ross and Schroeder 2014, 267), we would face the question of what MacFarlane calls "dual control": Whether the defaults are in charge or are to be departed from would depend on the risks and stakes of the situation, as captured by credences and utilities. But then it seems like credence-based cognition has to constantly supervise the simplified reasoning. The required computations would not be reduced, but rather increased, if the two are thought to run in parallel in this way (MacFarlane 2025, 851).

What these two have in common is that they inevitably lead to "belief" and "credence" competing for the same theoretical task—and for a task at which "credence" would inevitably be simply better. If credence reductionism is to be resisted, then, belief needs to find a job "in a different line of work altogether" (MacFarlane 2025, 851); one which makes use of belief's distinctive skill set.

What is this distinctive skill set? MacFarlane identifies two qualifications in particular which "belief" brings to the table and which "credence" cannot match:

First, belief has truth as its correctness condition. If p turns out not to be true, then belief that p was wrong. Whereas even arbitrarily high credence in p may not be proven wrong by p being shown not to obtain—even credence 1 does not, strictly speaking, rule out $\neg p$. As Williamson (2020) points out with the example of a (countably) infinite series of coin tosses: we certainly should have credence 1 in the proposition that at least one of them will come up tails[3]; and this credence will not have shown to have been wrong if—which is possible—not a single toss does come up tails.[4]

Second, belief—unlike credence—can serve as the basis for certain reactive attitudes, as described by Buchak (2014) : I cannot resent you for stealing my phone unless I believe you did. Credence, even an arbitrarily high one, is not enough.

With these talents, belief shouldn't try to compete with credence in explaining why individuals choose one permitted action over another. They qualify it for a different job: ascribing beliefs to individuals is crucial for the description of another realm of human activity, namely the social practice of reasoning (MacFarlane 2025, 855).

We ascribe to an agent the belief that p if p is for them a candidate reason—if, under suitable circumstances, they would assert p in the language game of giving and asking for reason and thereby assume responsibility for its truth, in the sense of: having reasons for thinking the sentence is true and being willing to provide these reasons upon request.

Believing is crucially intersubjective, and bound up with reasoning about the world in language, in a way in which having a credence is not. I want to illustrate this point from two directions: first, in the way MacFarlane does, who provides broadly cog-sci arguments as to why intersubjective communi-

[3]Assuming that for each coin toss, we have credence 1/2 that it comes up "heads" and 1/2 that it comes up "tails", and that the coin tosses are independent, probabilism commits us to assigning credence $\frac{1}{2}^n$ to the event "the first n coin tosses all come up 'heads'". As n approaches infinity, this credence will therefore approach 0 and the credence that at least one of the first n tosses was "tails" 1.

[4]Or consider selecting at random (sampling from a uniform distribution) a real number in the interval $[0, 1]$, and the event "the selected number is rational". There are rational numbers in this interval; in this sense the event is possible. But the set of rational numbers in $[0, 1]$ has measure zero; the probability of picking one is therefore 0.

cation requires a coarser grained attitude (compared with an individual's internal credal state). And then, in a more speculative philosophical way, drawing on Donald Davidson, who writes about the connection between participating in a practice of linguistic communication about what is true and the very availability of the concept of objectivity.

First, the MacFarlanian point: the norms of credence, which for him are just the synchronic and diachronic Bayesian requirements of probabilism and conditionalization, "demand an equilibrium of a very large number of mental states". He theorises that "[p]resumably we can attain this, at least approximately, in our own thought, because we have subpersonal mechanisms that are sensitive to irrationality" (MacFarlane 2025, 858). But in coordinating with others, in settling on a shared picture of the world and a shared sense of what there are and aren't reasons for, "everything has to be under personal-level control" (ibid.). He says to "think of belief and reasons as digital, in contrast to the analog distinctions involved in credence and rationality" (ibid.). The digital version of an analog signal loses some detail, but what it gains is the ability to be communicated efficiently and reliably—"a digital signal can be copied exactly, whereas an analog signal cannot". In line with this picture, on which the purpose of reasoning is to, in some sense, do a slightly worse version of what an individual's brain does sub-personally, MacFarlane views solitary reasoning as derivative of the more primordial practice of social reasoning: He cites Mercier and Sperber (2017), who write about their experimental results as confirming that people reasoning on their own do so "anticipating a dialogical context, and mostly to find arguments that support their opinion" (Mercier and Sperber 2017, 61).[5]

In a certain sense, MacFarlane does here end up saying that belief simplifies in a suitable way, just like Staffel (2019) or Ross and Schroeder (2014), whose views he rejected earlier; but while e.g. Staffel (2019) points to some complexity-theoretic results like Cooper (1990)'s[6] to argue that the individual's brain cannot be expected to perform the computations required for updating the individual's credences, MacFarlane's argument seems to rather point in the direction of information-theoretic results like the noisy channel theorem, and the fact that, with in a conversation, the number of premises that can be considered and whose reason-relation on a question under consideration can be taken into account is strictly limited in a way that an individual brain's holistic updating procedure need not be.

Donald Davidson (1982), in a much-hated paper which I, however, love, even though it is, objectively, a little bit of a mess, writes on language being required for the having of beliefs. Core to the concept of

[5]Note that this is compatible with something like the following going on: On a sub-personal level, an individual's brain constantly updates its credence distribution. Whenever an individual is (internally) reasoning in the non-probabilistic way—operating with a small set of premises, weighing whether they support an affirmative judgment on some proposition being considered—they are playing in their head through a potential conversation they might be having with other reasoners. This process may even be influenced through some mechanism (which we are not concerned with here and thus leaving unspecified) by the sub-personal credence distribution represented in the individual's brain.

[6]Which she reports as "probabilistic inference, including conditionalization, is NP-hard". What Cooper (1990) in fact shows is that general probabilistic inference is NP hard if the probability distribution is represented in the form of a Bayesian network. She does not cite any evidence to the effect that the human brain does in fact represent its credence distribution in this particular way.

belief, he argues, is that it is a state of an entity that can be true or false, correct or incorrect (it differs, I might add, from credence in that way). Essential to the concept of belief is the contrast between what is subjective and what is objective (or, which to Davidson is the same, intersubjective). Any being engaged in linguistic communication has this command of the subjective-intersubjective contrast (and therefore of the concept of truth): for having a conversation with another person entails having a concept of a shared world which both conversation partners are talking about. To understand the other person's utterances, I must think of her world as being the same as my world. Communication requires that my sentences and the sentences of my interlocutor have the same subject matter; and disagreement is only possible against a backdrop of many shared beliefs about that subject matter. Having a concept of such a shared world is the same as having a concept of truth and falsity of the internal states that correspond to one's linguistic utterances, since a belief that p is true iff p (iff p is the case in the world; and, in Wittgenstein's words, if this world were only my world, "whatever is going to seem correct to me is correct. And that only means that here we can't talk about 'correct'" (Wittgenstein 2009, sec. 258)). To illustrate the situation of a being which does not participate in such an activity Davidson paints the following picture: imagine you are fixed in place and unable to move. It is thus impossible for you to determine the distance of many objects. You know on which line through yourself they are, but not at what distance—it would be impossible to distinguish between certain small objects which are close by and large objects which are far away. Not being fixed in place, you are able to triangulate. From the difference in the way an object appears to you when you move a bit to the left or the right, you can infer how far it is from you. Now, he suggests that our sense of objectivity is itself an effect of an ability to triangulate; not triangulating a spatial object qua spatial object by moving in space, but triangulating an object qua object by accessing another subject's perspective on it by talking about it with them. The very distinction of thing-for-me and thing-for-everyone (i.e. object in the proper sense) is only opened up by interacting with another subject in a particular way: communicating with them about the object in language. (Davidson 1982, 105)

It is this activity, the activity which human beings engage in with each other to construct a shared objective world about which true and false things can be said, that makes the addition of the concept of "belief" to our psychological vocabulary necessary: while the behaviour of human beings as utility maximizers can be described and explained using credence alone, the behaviour of human beings as social reasoners can not. The making of an assertion is an assumption of responsibility for the truth of the asserted content (where this responsibility includes having reasons for thinking the sentence is true and being willing to provide and discuss them); and we use the concept of "belief" to describe and explain what kind of assertions someone is disposed to make.

Based on this account of belief, we can also explain why belief, unlike credence can serve as the basis for reactive attitudes: What certain reactive attitudes (like resentment, blame, pride or shame) require is precisely a reason. Unlike some other moods (like sadness), for these attitudes it always makes sense to ask (a version of): "Why do you feel proud?" / "Why do you blame Jones?"—and it would be incoherent to reply something along the lines of: "No reason, I'm just feeling proud" (MacFarlane 2025, 856).

2.3 The Respective Normativities of Belief and Credence

The above is an outline of when we should ascribe beliefs and credences to an agent; it doesn't yet say anything about what beliefs or credences an agent should have, or, conversely, under which circumstances an agent can be criticised as unreasonable for the beliefs she holds or as irrational* for her credences.

In line with our basic methodological behaviourism, we have defined these terms with reference to the kinds of behaviours which are their characteristic expression; action guided by decision theory for credence, and moves in the language game of giving and asking for reasons for belief. For each of these areas of human activity, there is a pre-existing normativity governing them. I will argue that the internal states of belief and credence inherit the normative status of the actions which are their characteristic expression: Since credence just is that internal state which makes one choose certain actions in certain situations, we will call it irrational* if the actions it makes the agent choose are irrational*. Likewise for belief: Since to believe q is to treat q as a candidate reason in conversation, believing q is irrational if it unreasonable to treat q as a candidate reason.

The paradigm example of an irrational* action I will consider to be: Entering into a set of bets which the agent recognises to jointly guarantee a loss of utility. It can never be rational* to give away utility for nothing, if keeping it is also permitted. Consequently, a set of credences is in any case irrational* if it makes one susceptible to a Dutch Book.

The language game of giving and asking for reasons includes a number of norms; for the application of the framework I am about to discuss, one is particularly relevant: I assume that it includes a norm like: "If you asserted p yesterday, and had good reasons for it, and have not become aware of any new reasons bearing on the truth of p; then you are still committed to p."

2.4 Application: Sleeping Beauty

[Write the whole sleeping beauty thing here]

2.5 In Search of A Pineal Gland

MacFarlane, in summarising the division of labour he has in mind between the concepts of "belief" and "credence", emphasises the relative independence between rationality and reasoning:

> Reasoning is the process of producing, assessing, and criticizing reasons. So understood, reasoning has only a tenuous relation to rationality. Rationality, as I understand it, is a matter of the internal coherence of one's attitudes, both synchronically and diachronically. This is what Bayesian epistemology and decision theory attempt to give an account of. One can be rational without ever reasoning, and without ever regarding anything as a reason. Conversely, reasoning—even good reasoning—does not necessarily make one rational. (MacFarlane 2025, 855)

Putting it this way, we are of course presented with the question—what is reasoning good for? MacFarlane's answer: "Coordinating with others" (MacFarlane 2025, 863) which, as discussed above, requires suitable coarse-graining.

But part of the story is missing: How does the coordinating affect the individual's actions? MacFarlane seems to be in a precarious position here: to the extent that the result of the communication is the formation, in the individual, of new beliefs, these beliefs should never feature in an explanation of that individual's actions—that is credence's job! And belief should not interfere with it, since it's talents do not qualify it for this job. But a story of how beliefs affect credences is lacking—and in fact the above quote seems to suggest that, in general, they do not.

Well, somehow they have to. There needs to be a pineal gland which connects the realm of credence and the realm of belief, on pain of casting the whole practice of reasoning as an epiphenomenon.

In this section, I want to suggest such a "pineal gland", a point where belief and credence do meet; I want to suggest that they do in the practice of betting. The result will be a view where reasoning is in a sense primary: certain choices of an agent will be seen as a reason to call that agent irrational, and irrational is something one should not be; thus reducing the normativity governing credences to the power of belief-based reasoning to determine what one should and shouldn't be or do.

A natural point to start when looking for a point where beliefs affect credences is conditionalisation, which may be framed as saying: If you have the belief that p, set your credence in q to $Cr(q \mid p)$. A drawback of just running with this is that it would seem to entail that one ought to have credence 1 or 0 in anything which one has a belief about (since probabilism requires $Cr(p \mid p) = 1$)—a view which is widely rejected, and, I think, for good reasons. It seems perfectly fine to assert something without thereby committing oneself to betting on it at any odds.

[…some kind of transition…]

Successful execution of a bet between to people from start to finish involves and connects both decision theory and reasoning, both credence and belief (I take general latent awareness of this fact to be the reason why Dutch Book arguments are often used to convince people of Bayesianism, and, as I will expand on later, are in fact the only correct justification of it). An individual's action within such a process is guided, at times, both by decision theory and by the normative requirements to which one is subject as a reasoner (and thus, believer). The way these are connected through a (real or imagined) bet will impose indirect epistemic requirements on credence, even if the only direct normative requirements on credence are pragmatic.

Here, I understand "betting" as the social practice where two parties belonging to the same linguistic community enter into an agreement with the content that an exchange of a set amount money should take place—usually at a future point in time which is at least loosely specified—conditional on the truth of a certain statement. This can (but doesn't have to) take the form of one party selling the other a "bet" in the sense defined in the introduction.

For a specific q, any such agreement is a bet on q iff the parties share an understanding that the pot should be awarded to the person who bet on q iff q. By entering into a bet on q, I commit myself to the this proposition about who should receive the pot under which conditions. When it is time to settle the bet, in order to settle it, the parties must reach agreement on whether q; that is, they must reach a state where both hold the same belief about q.

Based on this understanding of a bet, I want to propose what I call the Betting Norm, which I take to be more fundamental than the "Dutch Book" norm (and, in fact, explaining it). It concerns the situation where an agent is offered a bet, or a set of bets, and it specifies one way in which accepting such a (set of) bet(s) can be irrational* for her; namely, if she already expects that any reasonable person will in the end reach the judgment that the bets are to be settled in a way which would entail that she overall loses money.

> Betting Norm: Assume you are invited to enter into a (set of) bet(s). It is irrational* for you to accept, if you already believe for some ϕ that ϕ will be the only reasonable thing to believe at the time when the bet(s) are to be settled, and ϕ being true implies that the total payout of the (set of) bet(s) should be such that you lose money.

[…spell out how Dutch Books are of the form "if $\top$, you lose money", so that even though you do not know which contingent proposition will be the only reasonable one to believe, you do know that it will be untenable to deny $\top$, so that the Betting Norm explains why you should not enter into Dutch books…]

[… spell out how this motivates conditionalisation without committing one to the above view that credences in believed propositions have to be 1; this will work via the distinction between "I believe that p" and "I believe that p will be the only reasonable thing to believe" (i hope)…]

- One way in which beliefs affect credences is conditionalization: Coming to believe that q should have the effect of your credence in p changing to what used to be $Cr(p \mid q)$.
  - Does this imply that I should have Credence 1 or 0 in every proposition about which I have

a belief? Because rationality* requires that $Cr(p \mid p) = 1$ and $Cr(p \mid \neg p) = 0$.

  - Getting a bit ahead of myself, but relatedly: Does my approach commit me to saying I should bet at arbitrary odds on any proposition which I do believe?
  - It seems that where MacFarlane pulled belief and credence too far apart, my "fix" is tying them too closely together
  - I guess a solution would be to require withholding belief about future contingents.
  - But what about bets about past events (but which happen to be unknown to both me and my betting partner).
  - Perhaps the distinction between "I believe p" and "I believe that p will be the only reasonable thing to believe once [time t arrives] or [we look at the decisive evidence]" is relevant here. I do believe the non-contingent fact about the future that either r or $\neg r$ will come to pass; and I believe that in each scenario, it will be either q or $\neg q$ which will be the only reasonable thing to believe; though I do not yet know which one it will be.

- I want to say that the way beliefs affect credences in conditionalization is derivative of the way the two attitudes intersect in the practice of betting.

- Successful execution of a bet from start to finish involves and connects both decision theory and reasoning, and thus credence and belief. An individual's action within such a process is guided by both forms of normativity, and in a way which makes clear how what we (anticipate to) believe will impose indirect epistemic requirements on credence even if the only direct normative requirements on credence are pragmatic. If my description of the practice of betting is right, then what I will call the "betting norm", can be seen as even more fundamentally constraining our credences than the "Dutch Book" norm which I have thus far been working with, and in fact explaining it: If you believe that ϕ will be the only reasonable thing to believe at the time when the bet is to be settled (where by this I mean: it will not be a tenable position to deny ϕ), and if ϕ being true implies that the payout on the bets you consider entering into should be such that you lose money: Then do not enter into these bets.

- The betting norm says:

- Bets with nature

3 Basing Reasons and Attitudes on Prediction Market Prices

[…under construction…]

4 Prediction Markets

4.1 A (Rough) Argument for: Credences Should (Sometimes) Mirror Market Prices of Bets

The different forms of the efficient market hypothesis provide reasons not to expect to be able to "beat" the market, i.e. not to expect to know better than the market what the value of an asset is — with the weak form implying that you shouldn't expect to be able to beat the market if all the information you have access to is historical information about the asset's price; the semi-strong form suggesting that unless you have access to non-public information, you should expect to lose money if you bet against the market and the strong form suggesting that you should always expect this.

Now assume you find yourself in a situation like this, where you do not expect yourself to be able to win a bet against the market, and where the asset in question is a bet on p. If your credence in p was anything other than the market price of the bet on p, on the above framework you would be committed to entering into a bet against the market. So, by modus tollens, you do not want your credence to be anything other than the market price of a bet on p.

4.2 (Rough) Argument(s) for: Belief Should Not Usually be Affected by Market Prices of Bets

On the other hand, does the fact that the market price of a bet on p is c have any normative pull on your doxastic state regarding p (i.e. on the question of whether to believe or disbelieve p)?

A special case is of course a proposition p which itself talks about, or stands in any logical relationship to, the market price of a bet on it[7]; in which case the answer is yes. But these seem so weird as to in general be irrelevant.

Here are some arguments for "no, the market price has no normative pull on my believing or disbelieving p", based on the two main features of belief [(MacFarlane 2025)] (and I) have been emphasising: - Correctness Condition: The market price comes about by individual's decisions to purchase bets. These are entirely based on these individuals' credences, which, as shown above, do not necessarily have anything to do with the proposition's truth. For example, the market price of a bet on $HEADS$ among participants of the "Sleeping Beauty" experiment would be 1/3; but we agreed that the correct belief about the chance of $HEADS$ cannot be 1/3. - Role in Reactive Attitudes: It doesn't seem to be the case that an arbitrarily high market price of a bet can ever justify a reactive attitude like resentment. Again, considering a hypothetical market whose participants are in the relevant kind of scenario—if,

[7] Such a p could be a proposition like "The market price of a bet on this proposition at time t will be c." or "There is a market for bets on this proposition." But such self-referential markets would be strange enough as to wonder whether, with regard to the rationality of their price, all bets are off anyway.

in a large room, everyone has a rationally (arbitrarily) high credence in "Jake stole the phone", without anyone believing that Jake did it, because there are no appropriate reasons available for believing this; and if a prediction market was run in this room on "Did Jake steal the phone?", then the market price would be equal to the median credence, which, by stipulation, is arbitrarily high. Yet if anyone formed a belief that Jake did it on the basis of this price, they would do so without any reason known to anyone that would be suitable to justify such a belief. - Overall, the market price is problematically inert in reasoning: It does not talk or justify, does not participate in the social practice of reasoning. Where it may be justified to form a belief on expert testimony if one interprets such testimony as a kind of promissory note for reasons available upon request (and expert status as reason to expect these reasons to be good), no such reasons could be requested from the market—because, as the two above scenarios show, it is possible that no such reasons even exist or are known to anyone.

5 A Note on the Practical Implications of this View for Individual and Policy Decision Making

What does this mean for prediction markets as a potential source of information in an individual's, or a government's decision making? It is a consequence of this view that market prices of bets cannot legitimately be used for decisions which are matters of justice, or where anything other than simple maximisation of utility is at stake. But insofar as maximisation of utility is the overriding dimension along which a decision is to be evaluated, prediction markets are suitable for eliciting probabilities to be used in expected utility calculations.

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